



INGREDIENTS FOR A NERF ALL-GO RITHM



► We can efficiently sample from $K_\beta(x, dx')$ for all $\beta \in [0, 1]$ and $x \in \mathbb{X}$

► **Annealing path** of distribution $\pi_\beta(dx) = \pi_\beta(x)dx$ supported on X such that

► **Reference distribution** $\eta(dx) \equiv \eta(x)dx$ supported on $\mathbb{X}$ such that:

► **Target** distribution $\pi(dx) = \pi(x)dx$ supported ~~X~~ such that:

► For each β we can evaluate the un-normalised density $\gamma_{\beta}(x) \propto \pi_{\beta}(x)$

► **Local exploration** Markov kernels $K_\beta(x, dx')$ such that K_β is π_β -invariant for each β , i.e.

► We can efficiently sample and evaluate the normalised density $\eta(x)$ for all $x \in \mathcal{X}$

► We can evaluate the un-normalised density $\gamma(x) \propto \pi(x)$ for all x

▶ The path $\beta \mapsto \pi_\beta$ interpolating between the reference and target $\pi_0 = \eta$ and $\pi_1 = \pi$

► At $\beta = 0$ we have $K_\beta(x, dx')$ is the independent kernel for η



Announcing scheduling how to discontinue the announcing path

$$0 = \beta_0 < \beta_1 < \dots < \beta_N = 1$$

INGREDIENT FOR AN ANNEALING ALGORITHM

4

- ▶ **Target** distribution $\pi(\mathrm{d}x) = \pi(x)\mathrm{d}x$ supported $\mathbb{X}$ such that:
 - ▶ We we can evaluate the un-normalised density $\gamma(x) \propto \pi(x)$ for all x
- ▶ **Reference** distribution $\eta(\mathrm{d}x) = \eta(x)\mathrm{d}x$ supported on $\mathbb{X}$ such that:
 - ▶ We can efficiently sample and evaluate the normalised density $\eta(x)$ for all $x \in \mathbb{X}$
- ▶ **Annealing path** of distribution $\pi_\beta(\mathrm{d}x) = \pi_\beta(x)\mathrm{d}x$ supported on $\mathbb{X}$ such that
 - ▶ The path $\beta \mapsto \pi_\beta$ interpolating between the reference and target $\pi_0 = \eta$ and $\pi_1 = \pi$
 - ▶ For each β we we can evaulate the un-normalised density $\gamma_\beta(x) \propto \pi_\beta(x)$
- ▶ **Local exploration** Markov kernels $K_\beta(x, \mathrm{d}x')$ such that K_β is π_β -invariant for each β , i.e.
 - ▶ We can efficiently sample from $K_\beta(x, \mathrm{d}x')$ for all $\beta \in [0,1]$ and $x \in \mathbb{X}$
 - ▶ At $\beta = 0$ we have $K_\beta(x, \mathrm{d}x') = \eta(\mathrm{d}x')$ is the independent kernel for η
- ▶ **Annealing schedule** specifying how to discretise the annealing path

$$0 = \beta_0 < \beta_1 < \dots < \beta_N = 1$$

SEQUENTIAL VERSUS PARALLEL ANNEALING

- ▶ We simultaneously evolve a system of particles to approximation the annealing path